Grade 10 · Algebra II · Topic 2: Quadratic Functions and Equations · Lesson 2-4 — Overview of Complex Numbers

Choose Your Route

Three routes, same destination. Start anywhere — you may switch routes at any time, and you may try more than one.

Student name :	Date : .....

PRACTICE — Secure the method

Simplify, multiply, divide and solve

Items 1–3: simplify each expression. Items 4–6 are below.

1.	2.	3.
$\sqrt{-18}$	i^{13}	$(2+5i)+(3-2i)$

4. Multiply $(3+i)(2-4i)$ and write the answer in the form $a+bi$.
5. Divide $\dfrac{8-i}{2+i}$. Multiply top and bottom by the conjugate.
6. Solve $x^2+2x+5=0$ and write both solutions in the form $a+bi$.

Done when: every answer is in the form $a+bi$ and i^2 is replaced by -1 at every step.

APPLY — Use it in context

A circuit, a quadratic and a square root

1. Two parts in series have impedances $4 + 2i$ and $1 + 3i$ ohms (model values). With $V = 100$ V find the current $I = \dfrac{V}{Z}$ in the form $a + bi$ and its modulus.
2. Solve $x^2 - 10x + 29 = 0$ and write both solutions in the form $a + bi$.
3. Find real numbers a and b such that $(a + bi)^2 = -5 + 12i$.

Done when: the quotient is checked by multiplying back and the solutions are checked by substitution.

INVESTIGATE — Find out why

Arguments, not answers

1. Show that $(a + bi)(a - bi)$ is always a real number.
2. Is $\sqrt{-4} \cdot \sqrt{-9} = \sqrt{36}$? Explain which rule fails and why.
3. A quadratic with real coefficients has $b^2 - 4ac < 0$. Use the quadratic formula to explain why its two solutions are conjugates.

Done when: each argument uses $i^2 = -1$ in general terms, not only one numerical example.

Remember

These are names for the tasks, not names for you. Move between them whenever you want to. Everyone answers the same “Time to Check” at the end of the lesson.